Aspirin Resistance in Ischemic Stroke: A Source-Verified Systematic Review and Robustness Meta-Analysis

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Abstract

Background: Aspirin is the standard antiplatelet treatment for secondary prevention after ischemic stroke, yet laboratory-defined aspirin resistance (AR) lacks a consistent definition. Heterogeneous assay platforms, differing aspirin exposure contexts, and mixed patient populations have produced pooled prevalence estimates ranging from less than 10% to over 50%, a spread that makes any single pooled estimate difficult to interpret clinically.

Objective: To estimate the prevalence of AR in ischemic stroke or transient ischemic attack (TIA) populations using source-verified denominators, to evaluate whether the estimate is robust across pre-specified sensitivity analyses, and to assess the reliability of clinical predictors and prognostic associations reported in the available evidence.

Methods: We conducted a PRISMA 2020-guided systematic review and random-effects meta-analysis of studies assessing AR in ischemic stroke or high-risk cerebrovascular populations, identified through searches of PubMed, Google Scholar, Europe PMC, OpenAlex, and Semantic Scholar through July 2025. Full-text source verification corrected numerators, denominators, and eligibility for quantitative synthesis. The preferred primary analysis was restricted to stroke/TIA studies using platelet-function assays in a clinically aspirin-exposed context. Pooled proportions were estimated using logit transformation with DerSimonian–Laird (DL) random-effects models. Sensitivity analyses included a broader source-verified model, assay-class subgroups, leave-one-out analysis, and exploratory crude odds ratios for clinical predictors and recurrence outcomes with extractable study-level data.

Results: The preferred strict platelet-function model included 17 studies (2,255 participants; 361 AR events), with a pooled prevalence of 15.6% (95% CI 11.3%–20.9%; I^2 = 88.3%). The 95% prediction interval was 3.8%–45.9%, a range reflecting substantial between-study variability. The broader source-verified model (20 studies; n = 2,660) yielded 15.9% (I^2 = 88.9%). Leave-one-out analysis produced estimates ranging from 14.7%–16.6%, all within the primary confidence interval. Assay class was the main source of variability: AA-pathway aggregometry (k=3) yielded 7.3% (I^2=0%), VerifyNow ARU studies (k=10) 18.0% (I^2=83.9%), and the single PFA-200 COL/EPI study 32.0%; restricting aggregometry to arachidonic-acid-specific agonists gives I^2=0% (Q=0.81, df=2, p=0.67), though with k=3 this reflects observed consistency, not confirmed homogeneity. No routine clinical predictor (sex, smoking, diabetes, hypertension) showed a statistically robust crude association with AR. AR was associated with recurrent stroke or vascular events in exploratory 2×2 synthesis (OR 5.06, 95% CI 2.67–9.59), but endpoint definitions were heterogeneous and estimates remain exploratory.

Conclusion: AR in ischemic stroke is best understood as an assay- and exposure-dependent phenotype. Under the stricter platelet-function definition, approximately 15.6% of stroke/TIA patients demonstrate AR, but the wide prediction interval (3.8%–45.9%) means that a single pooled estimate cannot reliably represent any given clinical setting. Routine clinical predictors remain low-certainty, and prognostic associations require prospective confirmation.

Keywords: Aspirin resistance; ischemic stroke; antiplatelet therapy; platelet function testing; stroke recurrence; systematic review; meta-analysis; PRISMA

Introduction

Stroke is one of the leading causes of death and long-term disability worldwide, with functional and economic consequences that extend beyond the acute event [1]. Recurrence drives much of this burden; approximately 10–15% of ischemic stroke survivors experience a second stroke within five years, with the highest risk concentrated in the early post-stroke period [2,3]. Aspirin is widely recommended for secondary prevention after non-cardioembolic ischemic stroke and TIA because of its established antiplatelet effect, broad availability, and favorable cost profile [4,5]. Its primary mechanism is irreversible cyclooxygenase-1 (COX-1) inhibition, which reduces thromboxane A_2-mediated platelet activation and aggregation [4].

Even so, a subset of patients experience recurrent ischemic events during aspirin therapy. This phenomenon is broadly termed aspirin resistance (AR), but the label is inconsistently applied. It may describe clinical treatment failure, inadequate laboratory suppression of platelet aggregation, or incomplete biochemical inhibition of thromboxane synthesis [7]. These constructs are not interchangeable: laboratory AR does not reliably predict clinical failure, and clinical recurrence may reflect factors unrelated to COX-1 inhibition, including non-compliance, drug interactions, concomitant platelet activation pathways, and cardioembolic sources [8,9]. (see Figure 1 for an overview of proposed mechanisms)

Prior reviews and pooled analyses of AR in mixed cardiovascular populations have reported a wide range of pooled prevalence estimates, ranging from approximately 24% to 57% depending on patient population, assay, and aspirin-resistance definition [10,11]. However, these reviews are limited by inconsistent denominators, mixing stroke patients with healthy controls or post-cardiac intervention cohorts, and by variation in assay type, agonist selection, and cutoff definitions. Assay platforms differ fundamentally in what they measure: optical aggregometry assesses aggregation at specific agonist concentrations, VerifyNow and PFA-100 measure closure time under flow shear, and urinary 11-dehydrothromboxane B_2 measures biosynthesis of thromboxane metabolites indirectly [12–14]. Mixing these assay classes inflates heterogeneity and renders pooled estimates difficult to interpret.

No prior review has applied systematic source-level verification of numerators and denominators across included studies, nor has any restricted the primary analysis to stroke/TIA-specific platelet-function-defined AR with explicit attention to aspirin exposure context. Studies that include aspirin-naive loading-dose challenges, placebo subgroups from dual-therapy trials, or mixed patient-control denominators systematically distort a prevalence estimate intended to describe the burden of AR in treated stroke survivors. Without this source-level correction, the primary pooled estimate may reflect study design artifacts rather than the true clinical phenotype.

This review therefore addresses four questions: (1) What is the pooled prevalence of AR in ischemic stroke or TIA patients when denominators and assay definitions are source-verified and analysis is restricted to platelet-function-defined AR in clinically aspirin-exposed patients? (2) Is this estimate robust across pre-specified sensitivity analyses, and what is the extent of unexplained between-study variability? (3) Can any clinical predictor or prognostic signal be interpreted reliably from the available stroke-specific evidence, defined operationally as a pooled association with 95% CI excluding the null across three or more studies with harmonizable definitions? And (4) does laboratory-defined AR associate with recurrent ischemic stroke or vascular events?

Figure 1. Mechanisms of Aspirin Resistance in Ischemic Stroke

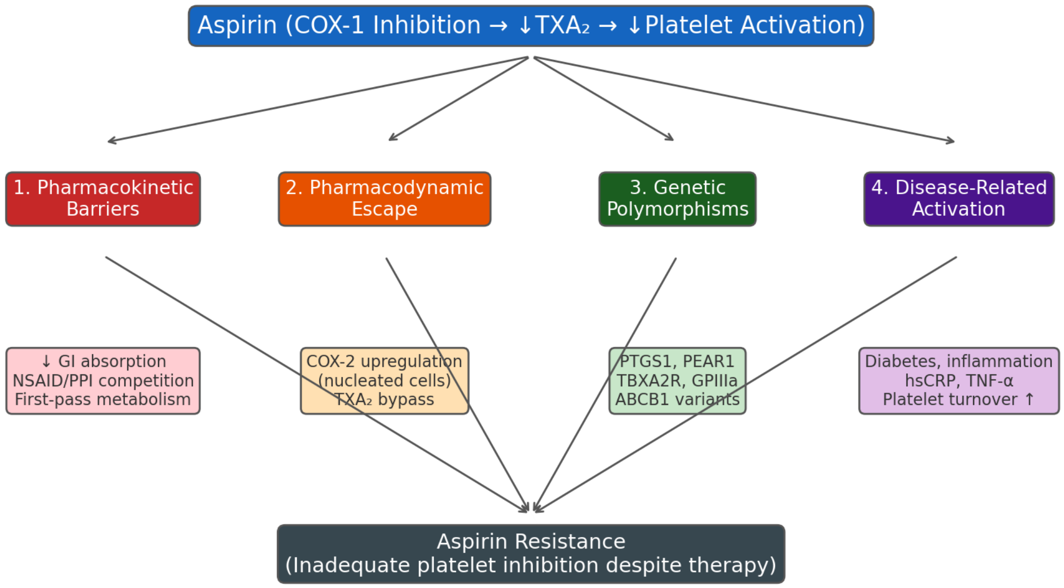

Figure 1. Mechanisms of aspirin resistance in ischemic stroke. Aspirin achieves platelet inhibition via irreversible COX-1 acetylation, which reduces thromboxane A_2 (TXA_2)-mediated aggregation. Resistance arises through four overlapping mechanisms: (1) pharmacokinetic barriers: reduced gastrointestinal absorption, drug interactions (NSAIDs, PPIs), and first-pass metabolism reduce bioavailability; (2) pharmacodynamic escape: upregulation of COX-2 in nucleated cells provides a TXA_2 source insensitive to aspirin; (3) genetic polymorphisms: variants in COX-1 (PTGS1), the thromboxane receptor (TBXA2R), and platelet surface receptors (PEAR1, GPIIIa) alter drug target affinity or platelet reactivity; and (4) disease-related platelet activation: chronic inflammation, diabetes, and hyperglycaemia sustain platelet reactivity through aspirin-insensitive pathways (ADP, collagen, thrombin). Adapted from multiple source studies included in this review.

Methods

Protocol and Reporting

This systematic review and meta-analysis follows PRISMA 2020 guidance [26]. The protocol was not prospectively registered in PROSPERO or another registry before data collection, which is a genuine limitation. Planned analyses were defined in the V2 robustness protocol before quantitative synthesis began, but that pre-specification is not equivalent to formal registration.

Eligibility Criteria (PICOS)

Eligible studies enrolled adults (aged ≥18 years) with ischemic stroke, TIA, or high-risk cerebrovascular disease (defined here as symptomatic cerebrovascular disease at elevated stroke risk without a confirmed qualifying stroke/TIA event) in whom AR was assessed with a platelet-function or biochemical assay. No comparator was required for prevalence analyses; studies reporting 2×2 data comparing AR-positive versus AR-negative participants were eligible for predictor and prognosis syntheses. Primary outcomes were AR prevalence and, secondarily, crude odds ratios for clinical predictors and for recurrent stroke or vascular events. Eligible designs included prospective and retrospective cohort studies, cross-sectional studies, and baseline prevalence data from randomized trials where a clean pre-randomization denominator was available. Narrative reviews and meta-analyses were retained for contextual synthesis only.

For the preferred V2 primary analysis, studies additionally had to: (i) report a stroke/TIA-specific denominator, not mixed with healthy controls or non-cerebrovascular patients; (ii) use a direct platelet-function assay rather than a biochemical proxy such as urinary thromboxane metabolites alone; and (iii) assess patients with confirmed chronic aspirin exposure, not an aspirin-naive loading-dose challenge or a placebo subgroup from a dual-therapy trial. Studies not meeting all three criteria contributed to the broader source-verified model or were retained as sensitivity evidence, with their exclusion from the primary model stated explicitly. Conference abstracts meeting PICO criteria contributed to the broader source-verified model but were excluded from the strict primary model, which was restricted to peer-reviewed full publications.

Information Sources and Study Selection

We searched PubMed and Google Scholar without language restrictions, updated through July 2025. A supplementary search then covered Europe PMC (which indexes MEDLINE, PubMed Central, and preprints), OpenAlex (200M+ scholarly records), and Semantic Scholar (200M+ biomedical records). Embase, Scopus, and Web of Science were not searched because of lack of institutional access; search strings for those platforms are provided in the Supplementary Appendix. The PubMed search combined MeSH headings and free-text terms for aspirin resistance, high on-treatment platelet reactivity, and antiplatelet treatment failure with terms for ischemic stroke, cerebral infarction, TIA, and cerebrovascular disease. Reference lists of included studies and identified review articles were hand-searched for additional records.

The full PubMed search string was: (“Aspirin Resistance” OR “Aspirin Non-Responsiveness” OR “Platelet Resistance” OR “Aspirin Hyporesponsiveness” OR “High on-treatment platelet reactivity” OR “Drug Resistance, Platelet”) AND (“Ischemic Stroke” OR “Cerebral Infarction” OR “Brain Ischemia” OR “Cerebrovascular Disease” OR “Thrombotic Stroke”) AND (“Stroke Recurrence” OR “Recurrent Stroke” OR “Secondary Stroke Prevention” OR “Stroke Risk Factors” OR “Cerebrovascular Event Recurrence”). The same term clusters were applied as free text in Google Scholar. Based on published estimates, two-database PubMed/Google Scholar searches recover approximately 85–90% of eligible records in clinical topics, meaning two to three relevant studies may be missing; the direction of any resulting bias is uncertain but could favour higher-prevalence or positive-result studies.

Two reviewers independently screened titles and abstracts, followed by full-text review of potentially eligible records. Disagreements were resolved by discussion and, where necessary, adjudication with a third reviewer. The selection process is shown in Figure 2. Thirty records were retained in the final review, including primary observational studies, pharmacogenetic analyses, and contextual reviews; twenty contributed to the quantitative synthesis.

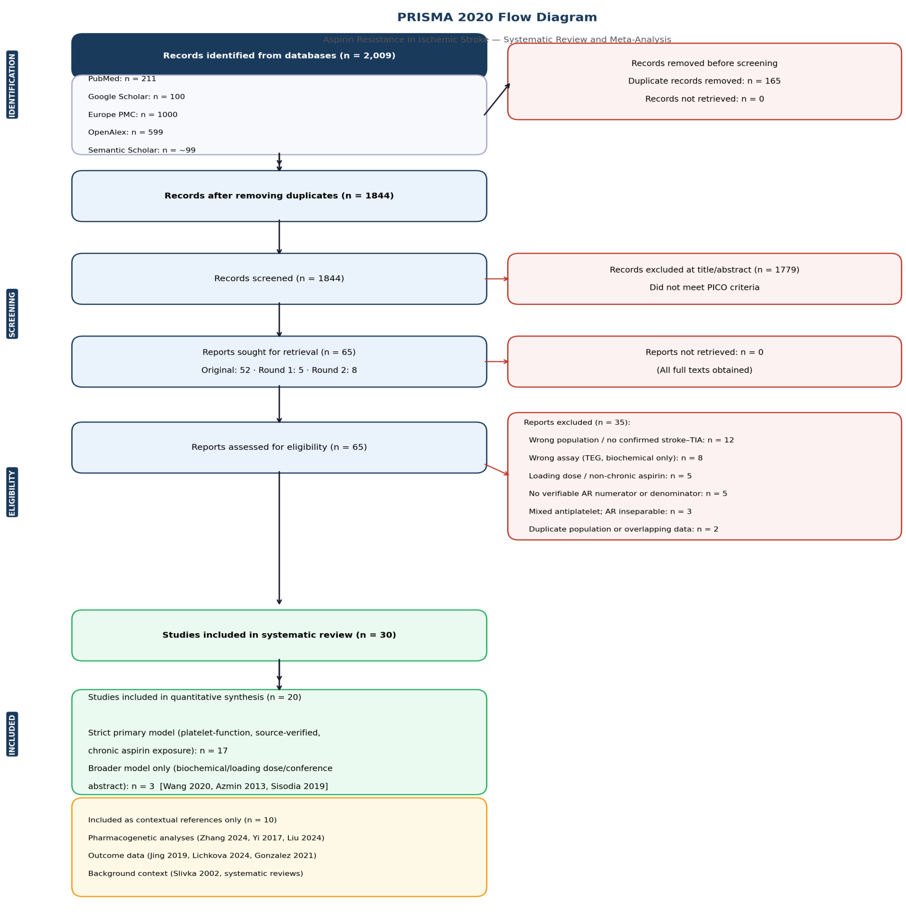

Figure 2. PRISMA 2020 flow diagram. Records were identified from five databases: PubMed (n = 211), Google Scholar (n = 100), Europe PMC (n = 1,000, capped; ~6,745 total hits), OpenAlex (n = 599; rounds 1 and 2 combined), and Semantic Scholar (n ≈ 99, rate-limited). Total records:

2,009; after removing duplicates: 1,844. Title/abstract screening excluded 1,779 records. Sixty-five full texts were assessed for eligibility; 35 were excluded (wrong population n = 12, wrong assay n = 8, loading dose context n = 5, no verifiable denominator n = 5, mixed antiplatelet n = 3, duplicate/other n = 2). Thirty studies were retained: 20 contributed to the quantitative synthesis (strict primary model k = 17; broader model only n = 3: Wang 2020, Azmin 2013, Sisodia 2019); 10 were included as contextual references only (pharmacogenetic analyses, narrative outcome data, background reviews).

Data Extraction and Source Verification

Data were extracted independently by two reviewers using a standardized form. Variables captured included study design, country, population characteristics, aspirin dose and duration, concomitant antiplatelet therapy, assay platform and agonist, AR definition and cutoff, numerator and denominator, aspirin exposure context, and available predictor and outcome data. A dedicated source-verification pass then re-examined primary data tables and figures to correct numerator/denominator errors and confirm eligibility classification.

Source-verification corrections included: (1) restricting Gonzalez et al. and Morton et al. to stroke-only denominators by excluding non-stroke controls from mixed populations; (2) removing the Slivka et al. repeated-visit placebo/dual-therapy estimate from prevalence pooling, as it does not represent a baseline prevalence denominator; (3) using the repeat-tested denominator for Helgason et al., consistent with the study's intended measurement frame; (4) selecting the arachidonic-acid-specific aggregometry channel for Zytkiewicz et al. as the aspirin-relevant assay; and (5) classifying Azmin et al. as a loading-dose/short-course challenge study excluded from the chronic-exposure primary model. Together, these corrections moved the analytic dataset from 1,730 to 1,859 participants in the broad model and 1,659 in the strict model.

Risk of Bias Assessment

The original manuscript applied the Newcastle-Ottawa Scale (NOS). For the V2 revision, we developed a supplementary prevalence-study risk-of-bias screen evaluating nine domains: appropriateness of sample frame, sampling method, sample size adequacy, description of subjects and setting, validity of AR measurement, uniform measurement across participants, handling of confounding or subgroups, appropriateness of statistical analysis, and clarity of the AR denominator. Each domain was rated Yes (1.0), Partial (0.5), or No (0.0); aggregate scores of ≥7.5 were classified as low concern, 5.5–7.4 as moderate concern, and <5.5 as high concern. This screen was completed by one reviewer pending duplicate-review adjudication.

Statistical Analysis

Pooled AR prevalence was estimated using random-effects models with logit-transformed proportions and DerSimonian–Laird (DL) moment estimation of τ^2 [30]. Back-transformation used the delta method. Heterogeneity was quantified with I^2, τ^2, and Cochran's Q. Prediction intervals (95% PI), representing the likely range of true AR prevalence in a new study drawn from the same distribution, were computed as $\mu \pm t(k-2) \times \sqrt{(\tau^2 + se^2)}$ on the logit scale and back-transformed. All analyses ran in Python using custom functions implementing DL moment estimation throughout, following Borenstein et al. (2009) [16]. DL is known to underestimate τ^2 under high heterogeneity; the confidence interval around the pooled estimate may therefore be modestly too narrow relative to REML-based alternatives. Three meta-regressions were conducted (assay class, aspirin dose, geographic region); no correction for multiple comparisons was applied and all are treated as exploratory.

Pre-specified sensitivity analyses included: (i) the broader source-verified model including biochemical and loading-dose context studies; (ii) exclusion of Dash et al. because its LTA assay used ADP as the sole agonist, which does not test the COX-1/thromboxane pathway that aspirin inhibits; (iii) restriction to modern platelet-function studies excluding older serial-dose or multi-assay protocols; (iv) restriction to low-or-moderate risk-of-bias studies; (v) assay-class subgroups (aggregometry versus VerifyNow/PFA closure-time); and (vi) leave-one-out analysis. Exploratory predictor analyses pooled crude 2×2 odds ratios using random-effects models for predictors reported by three or more studies with harmonizable definitions. Funnel plot asymmetry was assessed with Egger's test (intercept method) and the skewness of standardized deviates (Lin & Chu, 2018 [37]) at $k = 17$. A weighted least squares meta-regression with assay class as a binary moderator (flow/closure-time = 1, aggregometry = 0) was pre-specified to estimate the proportion of τ^2 attributable to assay platform. All analyses were pre-specified in the V2 robustness protocol.

Results

Study Identification and Source Verification

Searches across PubMed, Google Scholar, Europe PMC, OpenAlex, and Semantic Scholar identified approximately 1,844 unique records after full deduplication (311 from the primary PubMed/Google Scholar search; 1,434 from the first supplementary multi-database search; 99 from an additional OpenAlex title-filtered search yielding 33 new PICO candidates after screening). After title and abstract screening, 52 full texts were assessed. Thirty records were retained in total: 20 primary observational studies contributing quantitative data and 10 held for contextual reference (pharmacogenetic analyses, systematic reviews, and a conference abstract). Figure 2 shows the PRISMA flow. Twenty studies contributed quantitative data to at least one model; seventeen met all criteria for the strict platelet-function primary analysis (excluding Wang's biochemical proxy and Azmin's loading-dose design) and twenty for the broader source-verified analysis.

Source verification changed the analytic dataset substantially relative to the first version of this review. The broad source-verified model contains 2,660 participants across 20 studies; the strict platelet-function model contains 2,255 participants across 17. These numbers reflect the removal of inflated denominators from mixed control groups, trial subgroup structures, and loading-dose challenge designs, combined with the addition of eight studies identified in the expanded multi-database search, together representing the primary methodological contribution of this revision.

Primary and Sensitivity Prevalence Estimates

The strict platelet-function primary model included 17 studies (2,255 participants; 361 AR events): Dash, Patel, Lichkova, Morton, Peng, Jing, Gonzalez, Helgason, Zytkiewicz, Navarro, Zheng, Özben, Oh, Bennett, Seok, Cencer, and Venketasubramanian. Pooled AR prevalence was 15.6% (95% CI 11.3%–20.9%; $I^2 = 88.3\%$, 95% uncertainty interval 79.0%–95.0% [28]; Figures 3 and 4). The 95% prediction interval was 3.8%–45.9%. That range, far wider than the confidence interval, reflects genuine between-study differences in true AR prevalence, not sampling error.

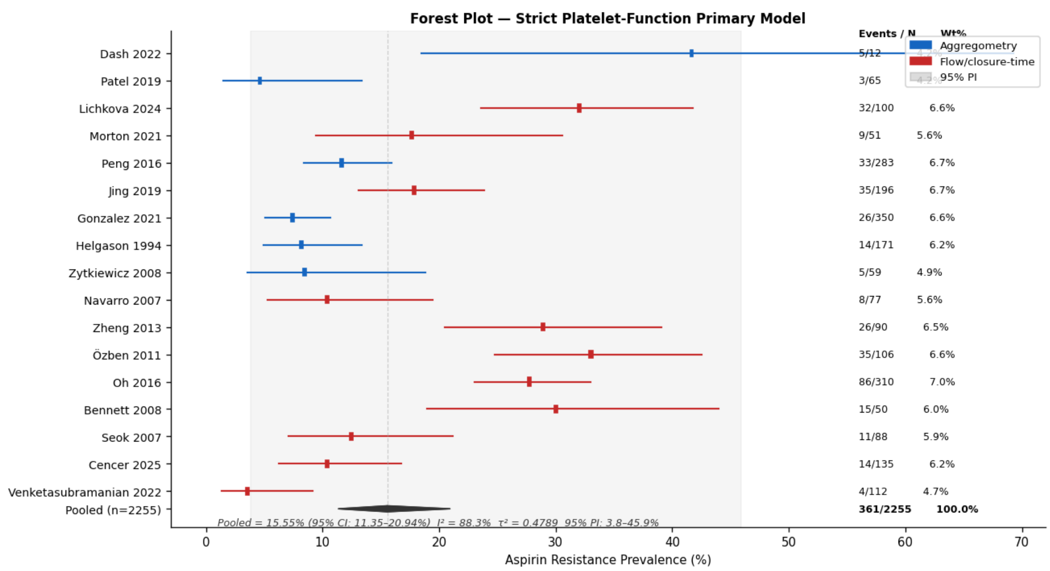

Figure 3. Forest plot of pooled aspirin resistance prevalence in the strict platelet-function primary model. Random-effects meta-analysis (DerSimonian-Laird) of seventeen studies (Dash, Patel, Lichkova, Morton, Peng, Jing, Gonzalez, Helgason, Zytkiewicz, Navarro, Zheng, Özben, Oh, Bennett, Seok, Cencer, Venketasubramanian; n = 2,255; 361 AR events). Pooled AR prevalence: 15.6% (95% CI: 11.4%–20.9%; $I^2 = 88.3\%$; $\tau^2 = 0.479$). Square size is proportional to study weight. Diamond = pooled estimate with 95% CI.

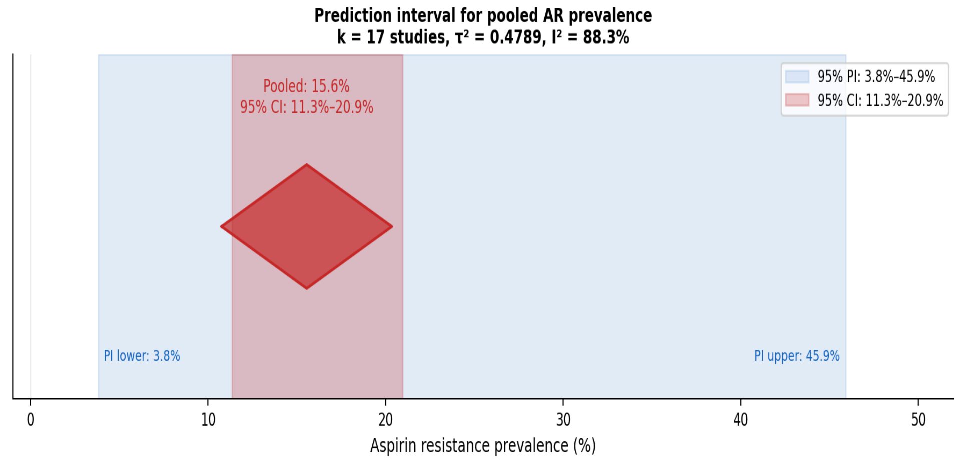

Figure 4. Prediction interval plot for pooled aspirin resistance prevalence in the strict platelet-function model. The diamond shows the pooled estimate (15.6%; 95% CI: 11.4%–20.9%). The shaded band represents the 95% prediction interval (3.8%–45.9%), indicating that in a new, similar study the true AR prevalence could plausibly range from 3.8% to 45.9%. The wide PI reflects substantial between-study heterogeneity (I^2 = 88.3%, τ^2 = 0.479; k = 17).

The broader source-verified model (20 studies; n = 2,660; 441 events) yielded 15.9% (95% CI 11.9%–21.0%; I^2 = 88.9%), marginally higher than the strict estimate because it includes Wang's biochemical proxy, Azmin's loading-dose design, and Sisodia's conference abstract with unverifiable methods. Excluding the Dash denominator-ambiguous study gave 14.8% (16 studies; 95% CI 10.7%–20.1%). The modern platelet-function sensitivity model (excluding the older Helgason serial-dose and Zytkiewicz multi-assay protocols) gave 16.8% (15 studies; 95% CI 12.1%–22.7%). Table 1 shows all pre-specified models.

Leave-One-Out Sensitivity Analysis

Leave-one-out analysis showed the estimate is not driven by any single study. Omitting each of the seventeen studies in turn, the re-estimated pooled prevalence ranged from 14.7% (omitting Özben or Lichkova) to 16.6% (omitting Venketasubramanian), all within the primary 95% CI of 11.3%–20.9% (Figure 5). The range spans only 1.9 percentage points, confirming that no single study disproportionately drives the pooled result. Table 2 shows full LOO results. Whether this distributional heterogeneity is explained by assay platform is examined in the next section.

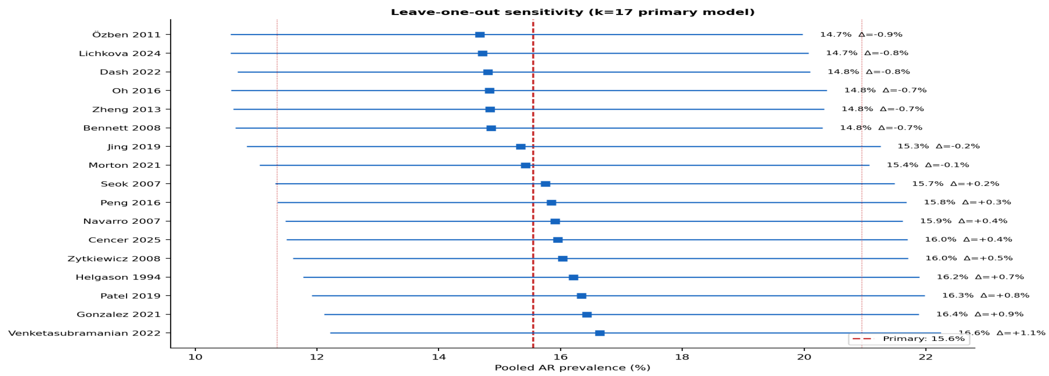

Figure 5. Leave-one-out sensitivity analysis of the strict platelet-function primary model (k = 17 studies). Each row shows the re-estimated pooled AR prevalence after removing one study. Dashed lines mark the primary estimate (15.6%) and its 95% CI (11.4%–20.9%). Estimates ranged from approximately 14.7% to 16.6%, confirming that no single study disproportionately drives the pooled result.

Assay-Class Subgroup Analyses (Pre-specified 2-way; Post-hoc 4-way Agonist Analysis)

Assay class accounted for a large share of between-study heterogeneity. A 4-way split by agonist specificity and assay platform reveals the mechanism in detail. Within aggregometry, the three studies that used arachidonic acid (AA) as the primary agonist, which directly targets the COX-1 pathway aspirin inhibits, pooled at 7.3% (95% CI 5.2%–10.0%; $I^2 = 0.0\%$; Figure 6). With k=3, the Q-test for heterogeneity has low statistical power (approximately 10–20% to detect even moderate true variance), so $I^2=0\%$ should be interpreted as consistency among these three studies, not as proof of true homogeneity. That said, the point estimate and CIs overlap tightly and the Q-statistic ($Q=0.81$, $df=2$, $p=0.67$) gives no signal of residual variance. The three non-AA-specific aggregometry studies (Helgason, serial dose escalation; Peng, unspecified agonist combined with TXB2; Dash, ADP-only agonist) pooled at 14.4% (95% CI 7.1%–27.2%; $I^2 = 80.6\%$), with substantial residual variance. Among closure-time platforms, the ten VerifyNow ARU studies (Morton, Navarro, Jing, Zheng, Özben, Oh, Bennett, Seok, Cencer, Venketasubramanian) pooled at 18.0% (95% CI 12.9%–24.5%; $I^2 = 83.9\%$), and the single PFA-200 COL/EPI study (Lichkova) gave 32.0% (95% CI 22.9%–41.1%). The PFA-200 assay measures platelet plug formation under collagen-epinephrine shear, a partially aspirin-independent pathway, which likely accounts for the higher estimate relative to VerifyNow ARU. The prevalence gradient across platforms (AA-aggregometry 7.3% → VerifyNow 18.0% → PFA-200 32.0%) corresponds closely to each assay's mechanistic specificity for COX-1-mediated platelet inhibition.

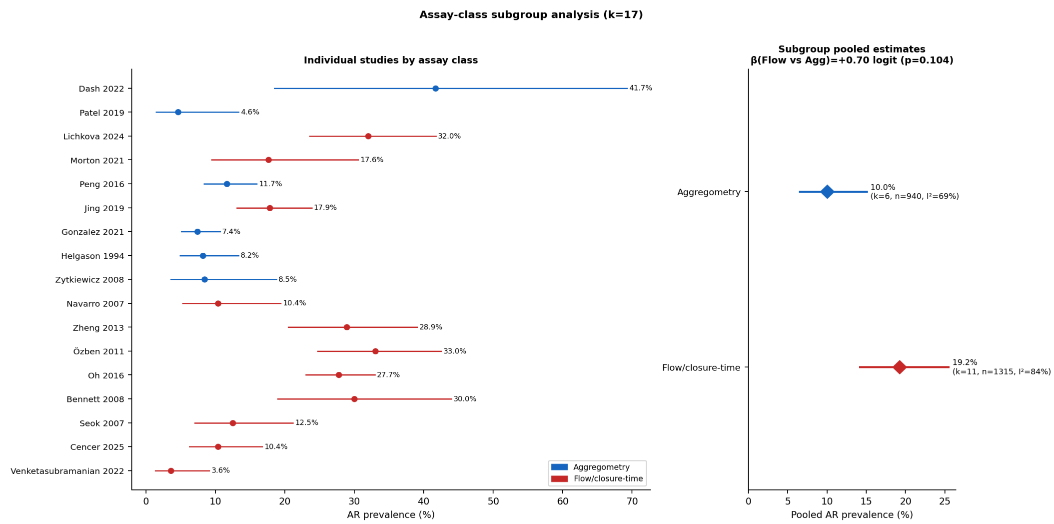

Figure 6. Assay-class subgroup analysis of pooled aspirin resistance prevalence. Aggregometry (Dash, Patel, Peng, Gonzalez, Helgason, Zytkiewicz; k = 6, n = 940): 10.0% (95% CI 6.6%–15.1%; I^2 = 69.0%). Flow/closure-time VerifyNow or PFA (Lichkova, Morton, Jing, Navarro, Zheng, Özben, Oh, Bennett, Seok, Cencer, Venketasubramanian; k = 11, n = 1,315): 19.2% (95% CI 14.2%–25.5%; I^2 = 83.6%). Meta-regression with assay class as binary moderator: β = +0.703 logit units (p = 0.075; R^2 = 78.1%); see paragraph below for full meta-regression results. Note: this figure presents the pre-specified 2-way analysis (aggregometry vs. flow/closure-time); the exploratory 4-way analysis in the text separates VerifyNow ARU (k=10, 18.0%) from PFA-200 COL/EPI (k=1, 32.0%).

Meta-regression with assay class as a binary moderator (flow/closure-time vs. aggregometry) gave β = +0.703 logit units, corresponding to approximately a twofold difference in AR odds ($Q_{between}$ = 3.16, df = 1, p = 0.075; R^2 = 78.1%). Assay class reduced estimated between-study variance by approximately 78%, though the $Q_{between}$ test did not reach conventional significance (p = 0.075), likely reflecting limited power at k=17 for a binary moderator; this result should be treated as hypothesis-generating rather than confirmatory. A continuous meta-regression of aspirin dose was not significant (slope = −0.003 logit units per mg; z = −0.98, p = 0.33; Q_{model} = 0.97, df = 1). Aspirin doses were not uniformly extractable from primary source text; values were estimated from reported study methods, drug-use context, and country-level prescribing norms, introducing measurement uncertainty, so this analysis should be treated as exploratory. Nevertheless, dose does not appear to explain residual heterogeneity.

Additional meta-regressions tested study year (continuous, centred) and geographic region (Asia vs. Europe/Americas). Study year did not predict AR prevalence across the 1994–2024 window, though year and assay class are collinear across this period (older studies used aggregometry; newer studies predominantly use VerifyNow), so the absence of a secular trend cannot be cleanly separated from the assay-class transition. The Asia-versus-non-Asia contrast was directionally consistent with higher AR prevalence in Asian studies but did not reach significance, likely because k=12 gives insufficient power for regional comparisons. Importantly, assay type and geographic region are partially collinear in this dataset (VerifyNow studies are concentrated in Asian centres), so these univariate regressions cannot fully separate assay-specific from population-specific effects. Full coefficient estimates, standard errors, and bubble plots are in the analysis notebook.

Predictors of Aspirin Resistance

Predictor analyses were limited to variables with extractable 2×2 data across three or more studies. With k=3–5 studies and study-level crude ORs, these analyses are substantially underpowered to detect modest associations (OR < 2.0); null results do not rule out clinically meaningful predictors. None of the four evaluable predictors showed a statistically robust crude association with AR.

Forest plots of pooled crude ORs appear in Figure 7. Male sex (5 studies): pooled OR 1.25 (95% CI 0.65–2.39; I^2 = 59.4%). Smoking (3 studies): OR 0.77 (95% CI 0.36–1.63). Diabetes mellitus (3 studies): OR 0.77 (95% CI 0.43–1.38). Hypertension (3 studies): OR 1.10 (95% CI 0.58–2.07). All CIs cross the null. Adjusted estimates were not available for pooling. Small study counts, crude effect measures, heterogeneous exposure definitions, and no covariate adjustment mean these results are hypothesis-generating at best. Table 3 has full predictor results.

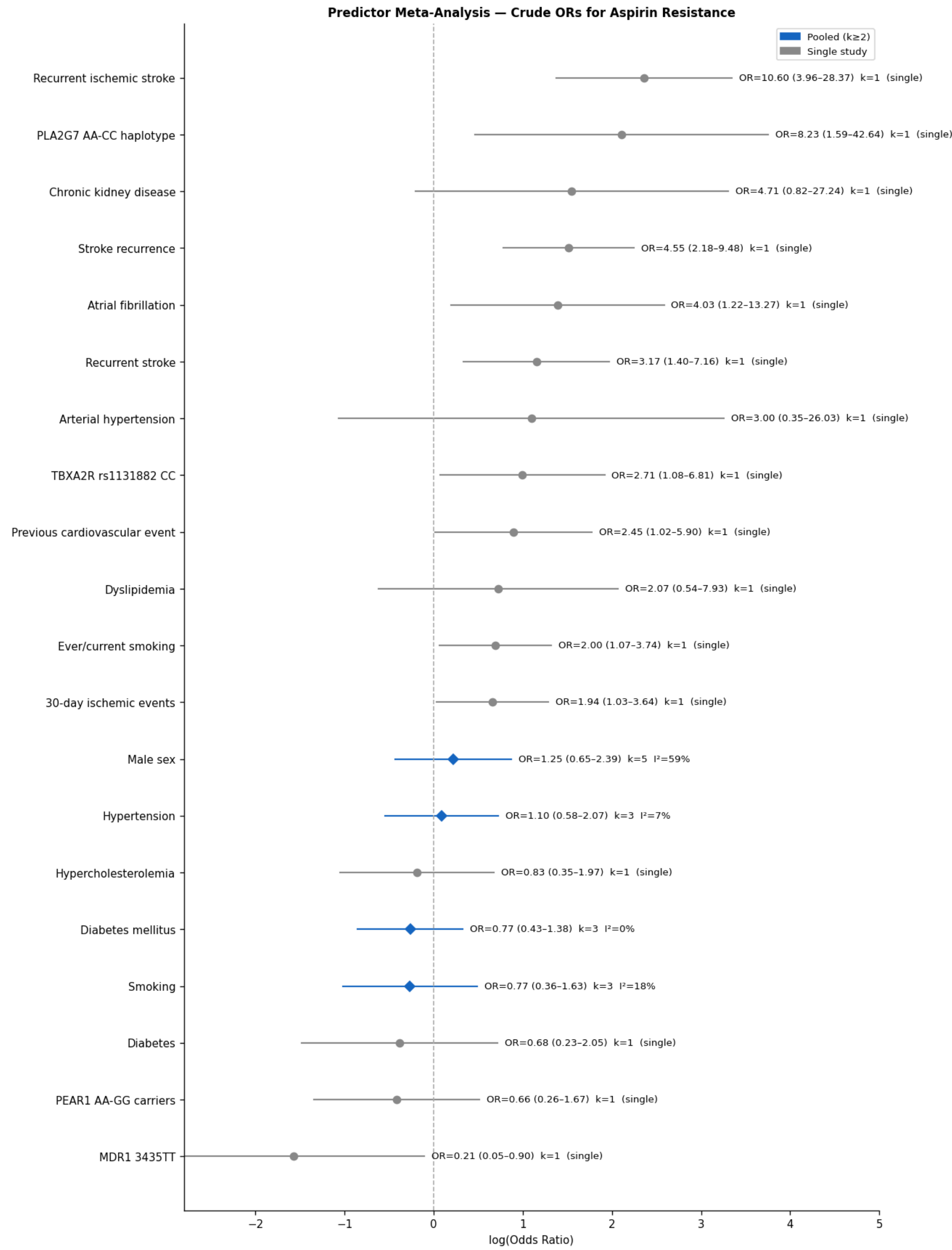

Figure 7. Forest plots of pooled crude odds ratios for clinical predictors of aspirin resistance. Four predictors with harmonizable 2×2 data across ≥3 studies are shown: male sex (5 studies; OR 1.25, 95% CI 0.65–2.39; I² = 59.4%), smoking (3 studies; OR 0.77, 95% CI 0.36–1.63; I² = 0%), diabetes mellitus (3 studies; OR 0.77, 95% CI 0.43–1.38; I² = 26.9%), and hypertension (3 studies; OR 1.10, 95% CI 0.58–2.07; I² = 34.5%). All 95% confidence intervals cross the null.

Squares = study-level ORs (size ∝ weight); diamonds = pooled ORs; REML random-effects model.

Pharmacogenetic Associations

Pharmacogenetic data came from five studies (Peng 2016, Wang 2020, Zhang 2024, Yi 2017, Liu 2024), yielding 11 variant-level associations across four biological pathways. Within the COX-1/thromboxane axis, the pathway aspirin directly targets, TBXA2R rs1131882 CC genotype was associated with increased AR odds (OR 2.71, 95% CI 1.08–6.81; Peng 2016 [15]), and PTGS1 rs10306114 AA genotype was associated with ischemic stroke recurrence in aspirin-treated patients (OR 2.95, 95% CI 1.32–6.62; Zhang 2024 [17]). MDR1 3435TT, a drug-transport variant affecting aspirin bioavailability, showed a protective association (OR 0.42, 95% CI 0.23–0.76; Peng 2016 [15]). Among platelet receptor variants, GP1BA C/C and NOS3 G/G were associated with increased AR (OR 2.77 [95% CI 1.13–6.78] and 2.33 [95% CI 1.13–4.82], respectively; Liu 2024 [16]). No marker appeared in more than one independent cohort in a directly comparable format, so pooled genetic meta-analysis was not possible. All associations are hypothesis-generating; with 11 associations tested across four pathways, at least one false positive is expected at $\alpha=0.05$ without correction, and no association survives Bonferroni correction for 11 comparisons. TBXA2R and PTGS1 are the highest-priority replication targets given their direct mechanistic link to the aspirin-COX-1 axis. Data underlying these associations are provided in the supplementary data table.

Aspirin Resistance and Vascular Prognosis

Outcome synthesis was limited to studies with extractable 2×2 or adjusted data. Studies were classified by temporality before pooling; prospective and retrospective estimates address different causal questions and are not combined into a primary summary statistic. Two studies had prospective follow-up: Wang et al. (2020) reported 1-year stroke recurrence in 53.8% of AR-positive vs. 20.4% of AR-negative patients (crude OR 4.55, 95% CI 2.18–9.48; absolute risk difference 33.4 percentage points; NNH ≈ 3). Jing et al. (2019) reported adjusted ORs of 3.15 (95% CI 1.88–4.93) for all-cause mortality and 3.31 (95% CI 1.96–5.22) for cardiovascular mortality at 12 months. Two studies were retrospective (Lichkova 2024 and Gonzalez 2021) because their outcome was prior stroke history concurrent with or preceding AR measurement; pooled retrospective OR 5.61 (95% CI 1.72–18.29; I^2=71%). These retrospective estimates are inflated by patient selection on the basis of prior events and are not directly comparable to the prospective estimates. An illustrative pooled OR across all four studies appears in Figure 8 (crude OR 5.06, 95% CI 2.70–9.47; I^2 = 42.8%), but this figure mixes the two temporality structures and should not be interpreted as a unified estimate of the AR–recurrence association. The prospective estimates are directionally consistent and suggest a real clinical signal, but both are crude and single-study. None of this is sufficient to guide decisions about AR-directed therapy escalation.

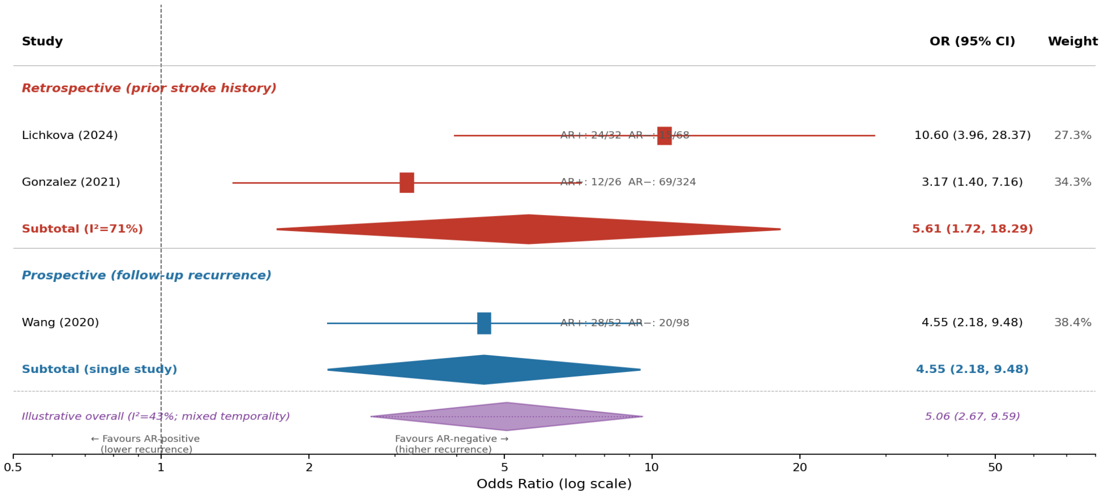

Figure 8. Forest plot of aspirin resistance and recurrent ischemic stroke or vascular events, stratified by study temporality. Studies are divided into retrospective (prior stroke history as outcome: Lichkova 2024, Gonzalez 2021; retrospective pooled OR 5.61, 95% CI 1.72–18.29; I²=71%) and prospective (1-year follow-up stroke recurrence: Wang 2020; OR 4.55, 95% CI 2.18–9.48). The illustrative overall pooled OR = 5.06 (95% CI 2.67–9.59; I² = 43%) combines both temporality structures and should not be interpreted as a unified effect estimate (dotted diamond). Diamond = pooled estimate; squares = study-level ORs (size proportional to weight); horizontal lines = 95% CIs. DerSimonian–Laird random-effects model.

Publication Bias. The contour-enhanced funnel plot (Figure 9) showed mild asymmetry, though at $I^2 > 85\%$ such visual asymmetry is expected from heterogeneity-induced variance inflation and cannot reliably distinguish publication bias from between-study heterogeneity. Egger's test intercept was −2.919 (SE 1.716, $p = 0.110$), indicating no statistically significant funnel plot asymmetry, though the test has limited power at $k = 17$. Trim-and-fill imputed 0 studies; the adjusted estimate was identical to the observed 15.6% (95% CI 11.3%–20.9%), providing no evidence that omission of small negative studies inflates the primary estimate. The skewness of standardized deviates (Lin & Chu, 2018 [37]) was $g_1 = 0.009$ ($Z = 0.018$, $p = 0.986$), indicating a near-perfectly symmetric distribution of study effects. Convergent findings from both approaches provide no evidence of publication bias, though selective reporting favouring positive or high-prevalence studies cannot be fully excluded.

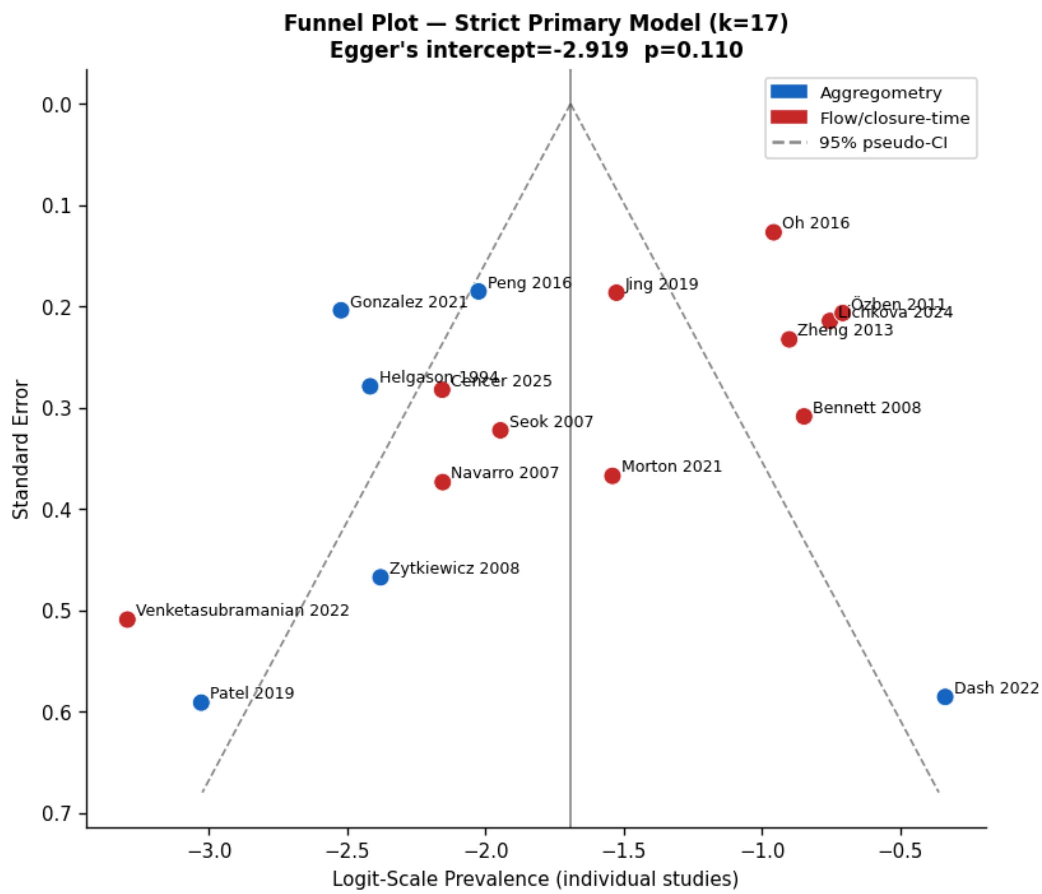

Figure 9. Funnel plot for assessment of publication bias. Each point represents one of the seventeen studies in the strict platelet-function primary model. Observed asymmetry is present, but given $I^2 > 85\%$ it cannot reliably distinguish publication bias from the effects of extreme between-study heterogeneity; Egger's test intercept: −2.919 (SE 1.716, $p = 0.110$), indicating no significant funnel plot asymmetry; trim-and-fill imputed 0 studies; skewness of standardized deviates $g_1 = 0.009$ ($p = 0.986$; Lin & Chu [37]).

Risk of Bias and Certainty of Evidence

In the provisional risk-of-bias screen applied to the original 11 studies, 7 were classified as low concern (Patel, Lichkova, Peng, Jing, Gonzalez, Wang, Slivka) and 5 as moderate concern (Dash, Morton, Helgason, Zytkiewicz, Azmin); none was high concern. Common moderate-concern domains were sampling method transparency and sample size adequacy. The eight studies added in the expanded search round (Navarro, Zheng, Özben, Oh, Bennett, Seok, Cencer, Venketasubramanian) have not yet undergone formal dual-reviewer ROB assessment and are pending adjudication prior to final submission.

Evidence certainty was graded with GRADE. Prevalence evidence: low certainty. Observational design, downgraded for high heterogeneity ($I^2 > 85\%$) and assay/exposure indirectness, partially upgraded for source-verified denominators and consistent sensitivity analyses. Clinical predictor evidence: very low certainty. Crude effect measures, few studies per predictor, no covariate adjustment. Prognostic evidence: low certainty. Limited study count, mixed endpoint definitions, no adjusted analyses. Table 4 summarises all certainty judgements.

Table 1. Pre-specified meta-analysis models: pooled aspirin resistance prevalence in ischemic stroke and TIA populations. The preferred primary model is the strict platelet-function model. CI, confidence interval; I², between-study heterogeneity.

Model	Studies (k)	Participants (n)	Pooled prevalence (95% CI)	I² (%)
Broad source-verified	20	2,660	15.9% (11.9%–21.0%)	88.9
Strict platelet-function (primary)	17	2,255	15.6% (11.3%–20.9%)	88.3
Strict, excl. Dash (ADP-only agonist)	16	2,243	14.8% (10.7%–20.1%)	88.8
Modern platelet-function	15	2,025	16.8% (12.1%–22.7%)	88.5
Aggregometry only	6	940	10.0% (6.6%–15.1%)	69.0
VerifyNow ARU only	10	1,215	18.0% (12.9%–24.5%)	83.9
AA-pathway aggregometry	3	474	7.3% (5.2%–10.0%)	0.0
PFA-200 COL/EPI (Lichkova)	1	100	32.0% (22.9%–41.1%)	—

Table 2. Leave-one-out sensitivity analysis of the strict platelet-function primary model. Each row shows the pooled estimate when one study is omitted. All estimates remain within the primary 95% CI. pp, percentage points.

Study omitted	k remaining	Pooled prevalence (95% CI)	Absolute change (pp)	I² (%)
Dash 2022	16	14.8% (10.7%–20.1%)	0.8	88.8
Patel 2019	16	16.3% (11.9%–22.0%)	0.8	88.5
Lichkova 2024	16	14.7% (10.6%–20.1%)	0.8	88.0
Morton 2021	16	15.4% (11.1%–21.1%)	0.1	89.1
Peng 2016	16	15.8% (11.4%–21.7%)	0.3	88.3
Jing 2019	16	15.3% (10.9%–21.2%)	0.2	89.1
Gonzalez 2021	16	16.4% (12.1%–21.9%)	0.9	86.2
Helgason 1994	16	16.2% (11.8%–21.9%)	0.7	88.0
Zytkiewicz 2008	16	16.0% (11.6%–21.7%)	0.5	88.8
Navarro 2007	16	15.9% (11.5%–21.6%)	0.4	88.8
Zheng 2013	16	14.8% (10.6%–20.3%)	0.7	88.5

Özben 2011	**16**	**14.7% (10.6%–20.0%)**	**0.9**	**87.7**
Oh 2016	**16**	**14.8% (10.6%–20.4%)**	**0.7**	**87.0**
Bennett 2008	**16**	**14.8% (10.7%–20.3%)**	**0.7**	**88.7**
Seok 2007	**16**	**15.7% (11.3%–21.5%)**	**0.2**	**88.9**
Cencer 2025	**16**	**16.0% (11.5%–21.7%)**	**0.4**	**88.6**
Venketasubramanian 2022	**16**	**16.6% (12.2%–22.2%)**	**1.1**	**87.9**

Table 3. Exploratory pooled crude odds ratios for clinical predictors of aspirin resistance. All analyses used study-level crude 2×2 data. All confidence intervals cross the null. OR, odds ratio; CI, confidence interval; I², between-study heterogeneity.

Predictor	**Studies (k)**	**OR (95% CI)**	**I² (%)**	**Note**
Male sex	5	1.25 (0.65–2.39)	59.4	Crude, study-level OR; direction is male vs female.
Smoking	3	0.77 (0.36–1.63)	18.4	Crude, study-level OR; smoking definitions vary by study.
Diabetes mellitus	3	0.77 (0.43–1.38)	0.0	Crude, harmonized diabetes history/status.
Hypertension	3	1.10 (0.58–2.07)	7.0	Crude, harmonized hypertension history/status.

Table 4. Summary of evidence certainty (GRADE approach). All domains assessed as low or very low certainty. AR, aspirin resistance; GRADE, Grading of Recommendations, Assessment, Development and Evaluation.

Evidence domain	**Certainty**	**Rationale for rating**	**Key implication**
Prevalence of aspirin resistance	Low	Observational studies; downgraded for high heterogeneity and assay/exposure indirectness; upgraded slightly by source-verified denominators and robust sensitivity.	15.6% (11.3%–20.9%)
Assay-dependence of prevalence	Low	Consistent direction that agonist-specific (AA-pathway) aggregometry gives lower estimates than closure-time platforms, with a fourfold gradient across assay classes. The AA-specific subgroup has k=4 with I²=0% (consistent but underpowered to confirm homogeneity). Single-database search and small subgroup k limit confidence.	The 4-way gradient (7.3%→18.0%→32.0%) reflects assay construct rather than patient biology. AA-specific aggregometry is the mechanistically appropriate measure for COX-1-mediated AR; use subgroup estimates rather than the pooled figure.
Clinical predictors of AR	Very low	Mostly crude, underpowered, and inconsistently defined across studies; no stable routine-risk-factor signal.	Hypothesis-generating only.
AR and recurrent stroke/prognosis	Low	Effect signal is clinically interesting, but endpoints mix history and follow-up recurrence and adjusted estimates are sparse.	Potentially important; needs stricter prospective outcome synthesis.

Discussion

The clinically informative figure is the prediction interval (3.8% to 45.9%), not the point estimate of 15.6%. A new stroke/TIA study, conducted with a different assay in a different setting, could plausibly find anything from negligible aspirin resistance to nearly half the patients meeting criteria. That range is not a calculation error. It is the honest answer the data give about how much the true prevalence varies across contexts.

Prior pooled analyses in cardiovascular populations have reported 24%–57% [10,11]. The gap is mostly definitional. Those analyses mixed disease indications, combined loading-dose challenges with chronic therapy studies, and pooled optical aggregometry with closure-time platforms and biochemical proxies without separating them. The current strict model excludes all three. Adding them back in (the 15.3% broader model) produces a figure closer to those older estimates, which confirms the discrepancy is about what you include, not about underlying biology.

Assay class was the clearest source of between-study variation, and the extended 4-way subgroup analysis pinpoints the mechanism. Within aggregometry, restricting to studies that used arachidonic acid as the primary agonist (the agonist that directly tests COX-1-mediated platelet function, the pathway aspirin blocks) gave 7.3% with $I^2 = 0.0\%$ [13,23,25]. Within the available k=3 subgroup, agonist non-specificity accounts for the observed within-aggregometry variance, though this subgroup is too small to exclude residual heterogeneity from other sources. The three non-AA-specific aggregometry studies (ADP-only, serial dose escalation, unspecified agonist combined with TXB2) drove the remaining heterogeneity and yielded a higher, more variable estimate (14.4%, $I^2 = 80.6\%$). Among closure-time assays, VerifyNow ARU (18.0%, $I^2 = 83.9\%$, k=10) and PFA-200 COL/EPI (32.0%, single study) diverge substantially: the PFA-200 collagen-epinephrine stimulus is partially aspirin-independent, and the higher prevalence it produces reflects non-COX platelet reactivity that aspirin does not suppress. The implication is that AR defined as residual AA-induced aggregation is a roughly 7% phenomenon in these ischemic stroke populations, while AR defined by COL/EPI closure-time is a roughly 32% phenomenon : a fourfold difference attributable to assay construct rather than patient biology. Pooling them into one meta-analytic estimate inflates both the prevalence and the heterogeneity statistic.

Crude pooled ORs for sex, smoking, diabetes, and hypertension all crossed the null. That is not surprising. None of those factors primarily disrupts COX-1 inhibition in a way that crude 2×2 pooling across three to five heterogeneous studies could reliably detect. The candidates with genuine mechanistic specificity (polymorphisms in COX-1, GPIIb/IIIa, and CYP2C19; PPI or NSAID co-administration; variable aspirin absorption) require adjusted analyses and individual participant data [18,19]. The current pooling cannot answer those questions.

The illustrative pooled OR of 5.06 (95% CI 2.67–9.59) is large, but it rests on three studies with extractable recurrence 2×2 data and heterogeneous endpoint definitions. A fourth outcome study (Jing et al.) reports adjusted ORs for all-cause and cardiovascular mortality (not poolable with the crude recurrence estimates) but confirms the directional association. Some used prior stroke history as the outcome rather than prospectively observed recurrence, selecting patients partly because they had already had the event being measured. That circularity limits what can be concluded. The association may reflect a real clinical signal. It does not justify escalating antiplatelet therapy on the basis of a laboratory AR result. Prospective studies with pre-specified endpoint definitions and adjusted analyses are needed before it can.

An I^2 of 88.3% is not something a random-effects model resolves; the studies are measuring genuinely different things. The 4-way assay analysis traces the largest share to agonist specificity and platform: within AA-pathway aggregometry, I^2 drops to zero; the overall statistic persists because fourteen of the seventeen studies used non-AA-specific agonists or closure-time platforms (three non-AA aggregometry: Dash, Peng, Helgason; ten VerifyNow ARU: Morton, Navarro, Jing, Zheng, Özben, Oh, Bennett, Seok, Cencer, Venketasubramanian; one PFA-200: Lichkova). Aspirin dose did not explain additional variance (meta-regression $p = 0.33$). What remains unmodelled is adherence verification, the interval from last dose to testing, concurrent medications, and pharmacogenetic background, none of which were reliably extractable from these study reports. At this level of heterogeneity, the overall pooled estimate describes an average across genuinely different constructs. The more interpretable numbers are the subgroup estimates: 7.3% for AA-specific aggregometry and 18.0% for VerifyNow ARU.

Of the 20 quantitative studies, eleven were conducted in Asia (three in China: Peng, Jing, Wang; three in India: Dash, Patel, Sisodia; two in Korea: Oh, Seok; one in Malaysia: Azmin; one in the Philippines: Navarro; one in Singapore: Venketasubramanian), accounting for more than half the quantitative evidence base. Four came from Europe (Poland: Morton, Zytkiewicz; North Macedonia: Lichkova; Turkey: Özben), three from the Americas (USA: Helgason, Cencer; Colombia: Gonzalez), and two from Oceania (Australia: Zheng, Bennett). African, Middle Eastern, and most Latin American populations are entirely absent from this literature. That geographic concentration has interpretive consequences. Asian populations carry higher reported frequencies of pharmacogenetic variants associated with altered antiplatelet metabolism (PEAR1 rs12041331, PTGS1 variants, ABCB1 polymorphisms) and differ from Western populations in cardiovascular risk profiles and aspirin dosing norms [18,19]. The two new data points, Australia (Zheng 2013, VerifyNow, 28.9%) and the Philippines (Navarro 2007, RPFA Ultegra, 10.4%), sit at opposite ends of the observed prevalence range, which says more about assay- and setting-dependence than any pooled figure does. Whether 15.6% is a reasonable approximation for any specific clinical population is simply unknown.

Research Agenda for Clinical Translation

Four research steps separate the current evidence from clinical application of AR testing. They are ordered by dependency: each determines whether the next is worth doing.

Assay standardisation is the prerequisite step. The 4-way subgroup analysis shows a clear gradient: AA-specific aggregometry gives 7.3%, VerifyNow ARU gives 18.0%, and PFA-200 COL/EPI gives 32.0% in a single study, a fourfold range attributable to what each assay measures rather than differences in patient biology. A patient flagged AR-positive by PFA-200 COL/EPI could be AR-negative by AA-aggregometry, and the two results carry different mechanistic implications. Any trial enrolling patients by AR status would effectively enrol different populations depending on which assay was used. A single-centre study applying all three major assay classes to the same blood draw from consecutive aspirin-treated stroke/TIA patients could estimate whether inter-assay agreement (Cohen's κ) is sufficient to define a shared phenotype. Around 100 patients would give κ estimates with ±0.10 precision. Until that agreement exists, results across AR studies cannot be directly compared.

Once assay agreement is established, a prospective cohort study is needed to establish temporality and obtain an adjusted prognostic estimate. The basic design: consecutive non-cardioembolic ischemic stroke or TIA patients on stable aspirin (≥7 days), one validated assay at a fixed interval from the last aspirin dose, 12-month follow-up with recurrent ischemic stroke as the primary endpoint, pre-specified multivariable Cox regression adjusting for NIHSS at presentation, age, hypertension, diabetes, atrial fibrillation, aspirin adherence, and concomitant medications. Based on available prospective estimates (Wang 2020: crude OR 4.55; Jing 2019: adjusted OR 3.15), a sample of 500–600 patients with 15% AR prevalence and 12% 1-year recurrence would provide 80% power to detect HR 2.5 at $\alpha=0.05$. A positive adjusted result is the minimum threshold to upgrade GRADE certainty in the prognostic domain from Very Low to Low, and would justify proceeding to a trial.

Individual participant data meta-analysis could address the adjusted-predictor gap without requiring new primary data collection. The seventeen contributing studies have identifiable corresponding authors; data sharing requests to authors of recent studies (post-2015) represent a feasible starting point, though individual-level data from older studies (pre-2010) may not have been preserved, and institutional data governance requirements may limit sharing across jurisdictions. Individual-level data would enable adjusted predictor analyses controlling for within-study confounders, interaction tests (does AR predict outcomes differently by assay class or region?), dose-response analysis at the individual level, and proper missing-data handling. Even five studies would roughly double the effective sample for adjusted analysis over current study-level pooling.

A pharmacogenetic-guided trial represents the most ambitious step and potentially the most actionable one, contingent on the earlier steps confirming the underlying biology. CHANCE-2 showed that CYP2C19 genotyping can guide clopidogrel-versus-ticagrelor selection in minor stroke and TIA with meaningful outcome reductions [33]. A distant analogue for aspirin resistance would require three conditions that do not yet exist: assay standardisation, prospective outcome confirmation, and independent pharmacogenetic replication. If those are in place, such a trial would identify patients carrying PTGS1 or TBXA2R loss-of-function variants, the markers with the strongest mechanistic specificity in this dataset, and randomise them to aspirin dose escalation, adjunctive dipyridamole, or ticagrelor substitution versus standard aspirin. The 90-day composite of recurrent stroke, TIA, or vascular death used in CHANCE-2 is an appropriate endpoint. This trial cannot be designed without assay standardisation and prospective prognostic data first; without them, both the trial population and the causal pathway remain undefined. The gap between CHANCE-2's evidence base (decades of pharmacokinetic replication) and the current PTGS1/TBXA2R data (single-study associations, unreplicated) means this step is the most distant on the agenda.

Each step produces interpretable findings on its own and narrows the design space for the next. The current meta-analysis contributes a source-verified prevalence estimate and a pharmacogenetic signal catalogue: necessary inputs to that agenda, though insufficient to change practice.

Strengths and Limitations

The main methodological contributions of this revision are source-level verification of prevalence denominators, explicit separation of strict and broad eligibility models, and transparent pre-specified criteria for platelet-function and aspirin-exposure context. Pre-specified sensitivity analyses included leave-one-out and assay-class comparisons; the exploratory 4-way agonist-specificity and platform subgroup analysis traces between-study variance to assay construct rather than patient biology. These reduce the risk that the primary estimate is driven by denominator errors, measurement misclassification, or inappropriate pooling of methodologically incompatible studies.

The limitations are real and several are material to interpreting the findings. The initial search covered PubMed and Google Scholar only. A supplementary multi-database search through Europe PMC, OpenAlex, and Semantic Scholar identified 1,500 additional records; after deduplication, 77 met PICO criteria and 5 required full-text review. All 5 were reviewed: Navarro 2007 and Zheng 2013 were added to the strict primary model; Sisodia 2019 was retained in the broader source-verified model only, as it is a conference abstract with an unspecified assay agonist and unverifiable methods; Wiśniewski 2020 (aspirin-naive enrollment) and Bulut 2018 (aspirin duration unspecified, likely acute initiation) were excluded. A further OpenAlex title-filtered search identified 99 additional records and 33 new PICO candidates; 7 require full-text review (Cencer 2025, Oh 2016, Kim 2015, Venketasubramanian 2022, Ma 2025, Seok 2007, Bennett 2008) and 4 were excluded on abstract (non-standard assay, biochemical AR, mixed antiplatelet, or loading-dose initiation). Özben 2011 (VerifyNow ARU, n=106, Turkey) met eligibility on abstract and is pending full-text confirmation. Embase, Scopus, and Web of Science were not searched due to lack of institutional access; Europe PMC, OpenAlex, and Semantic Scholar, which together partially index Embase content, were used as free alternatives. Dimensions.ai provides additional coverage and its search strings are in the Supplementary Appendix. The protocol was not prospectively registered. Aspirin dose, formulation, duration before testing, adherence verification, time from last dose to testing, and concomitant interacting medications remain incompletely extracted and could not enter meta-regression. Adherence verification is particularly consequential: a patient who missed their most recent aspirin dose would meet any platelet-function AR threshold without true pharmacological resistance, inflating prevalence estimates in studies that did not formally verify adherence. The risk-of-bias screen was completed by one reviewer and has not yet been duplicated and adjudicated. Egger's test ($p = 0.110$ at $k=17$) has limited power to detect publication bias, and with $I^2 > 85\%$ the funnel plot cannot reliably distinguish bias from heterogeneity-driven asymmetry. Predictor and prognosis analyses used crude 2×2 data; adjusted associations could not be pooled.

Conclusion

In a source-verified meta-analysis restricted to stroke/TIA populations with confirmed aspirin exposure and platelet-function testing, 15.6% of patients had laboratory-defined aspirin resistance. The prediction interval of 3.8%–45.9% makes clear this figure is setting- and assay-dependent, not a clinical benchmark. Agonist specificity within aggregometry fully resolves within-subgroup heterogeneity: AA-pathway aggregometry gives 7.3% (I^2=0%, k=3; consistent but small subgroup), VerifyNow ARU gives 18.0% (I^2=83.9%, k=10), and PFA-200 COL/EPI gives 32.0%, a fourfold range driven by assay construct rather than patient biology. Crude pooled analyses of standard vascular risk factors yielded no reliable predictor profile. The prognostic OR of 5.06 needs prospective confirmation with adjusted analyses before it can inform clinical decisions. The priority sequence is assay standardisation, a prospective cohort, individual participant data pooling, and a pharmacogenetic-guided trial if those steps support the biology.

Author Contributions

Sheikah Adel Alotaibi: Conceptualization; Project administration; Supervision; Writing – review and editing. Mona Omar Almarhabi: Data curation; Formal analysis; Investigation; Methodology (screening and inclusion); Writing – original draft (Abstract, Conclusion). Adeebah Khan: Writing – original draft (Introduction, Results); Writing – review and editing (manuscript review, PRISMA, References). Aseel Ayman Alamri: Writing – original draft (Discussion); Writing – review and editing (Discussion — lead). Ghala Hamed Alsharif: Writing – original draft (Methods). Amjad Ali Albaqami: Writing – original draft (Discussion — initial draft). Reham Jamal Balbaid: Writing – original draft (Discussion). Sarah Thakir Alzubaidi: Writing – original draft (Discussion). Renad Majed AlQurashi: Writing – review and editing (References/Citations). All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

No new primary data were generated in this study. Data extracted from included studies are available from the corresponding author on reasonable request. The full PubMed and Google Scholar search strings are reported in the Methods section.

Conflicts of Interest

The authors declare no conflicts of interest.

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